

Smart Campus Management System (SCMS)

- Capstone Project Documentation

1. Introduction

- **Project Title:** Smart Campus Management System (SCMS)
- **Target Platform:** ServiceNow Platform
- **Application Scope:** x_2065601_smart_0
- **Developer:** Santhosh Akkiseti

2. Project Overview & Executive Summary

The Smart Campus Management System is an enterprise-grade automation solution built on the ServiceNow platform to modernize university operations, eliminate fragmented administrative paperwork, and streamline campus workflows through a centralized self-service portal.

The platform is engineered within a secure, isolated scoped application container (x_2065601_smart_0) to guarantee absolute namespace segregation for all custom data tables, business logic, and UI elements. Navigation is organized through dedicated application menus and structured module category separators that cleanly segment high-level administrative management from everyday student and faculty interactions.

3. Architecture, Data Schema & Database Design

The backend data model relies on a robust relational schema featuring dedicated master tables for academic departments, courses, students, and faculty, alongside transactional tables designed to capture student course registrations, leave requests, and infrastructure maintenance complaints.

- **Master Entity Framework:** Core records are organized via structured dictionary attributes, form layouts, views, and list filters across custom tables (x_2065601_smart_0_department, x_2065601_smart_0_course, x_2065601_smart_0_student, and x_2065601_smart_0_faculty).
- **Transactional Operations:** Operational activities are tracked via dedicated mapping and request records, including student course enrollment tables, leave requests, and campus complaint logs.

4. System Automation, Logic & Business Rules

To ensure rigorous data integrity and seamless user experiences, the application employs multiple layers of platform automation:

- **Auto-Numbering & Identity:** Configured automatic record sequencing for streamlined tracking.
- **Client Scripts & UI Policies:** Dynamic form behaviors validate user input in real time, such as enforcing mandatory fields and executing date validation scripts.
- **Backend Logic & Utilities:** Reusable Script Includes ([scmsutils](#)) handle custom calculations, while advanced Business Rules prevent duplicate entries and automate background field updates and ticket routing rules.

5. Service Delivery, Workflows & Notifications

End-users interact with the platform via structured Service Catalogs and intuitive portal interfaces.

- **Flow Designer Automation:** Service catalog requests trigger automated background sequences managed by Flow Designer, processing student course registrations and campus complaints asynchronously without manual bottlenecks.
- **Notifications:** System-generated email alerts keep students and staff informed of status transitions, such as leave approvals or ticket updates.

6. Dashboards, Analytics & Application Deployment

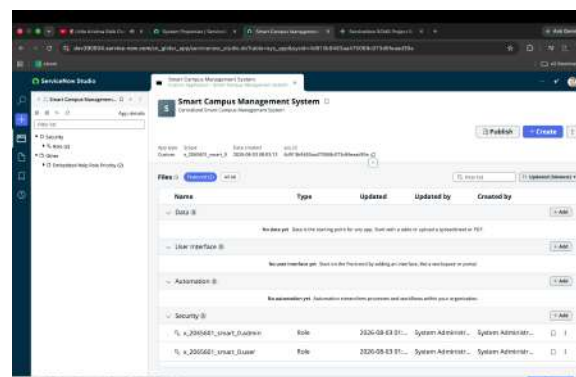
The platform provides comprehensive administrative oversight through real-time analytics and reporting:

- **Operational Visibility:** Administrators monitor key metrics via dedicated dashboards tracking complaint distributions by category, open leave request backlogs, campus overview metrics, and course enrollment trends.
- **Deployment & Packaging:** The entire application scope and configuration are successfully compiled and packaged into a standard ServiceNow Update Set for clean environment migration and export.

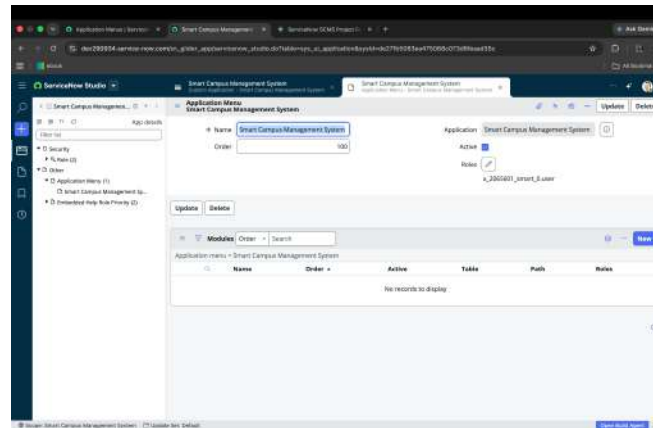
Phase 1: Project Initialization & Navigation Framework

- **Technical Report Matter:** This phase establishes the foundation of the smart campus environment by creating an isolated scoped application container to prevent data leakage and ensure namespace security. Following this, custom application menus and category separators are established to partition backend administration from front-end user navigation.

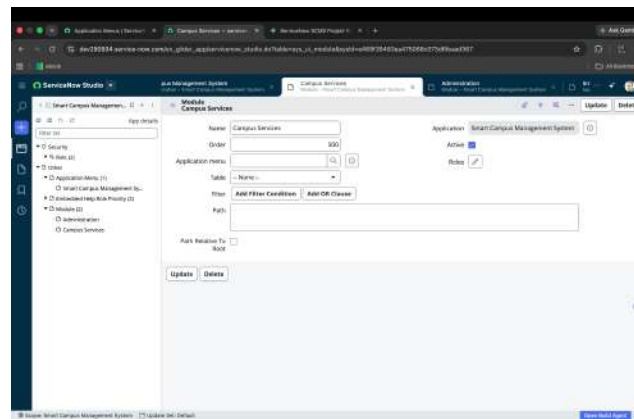
Figure 1.1: Scoped Application initialization.



- *Figure 1.2: Primary Application Menu configuration.*

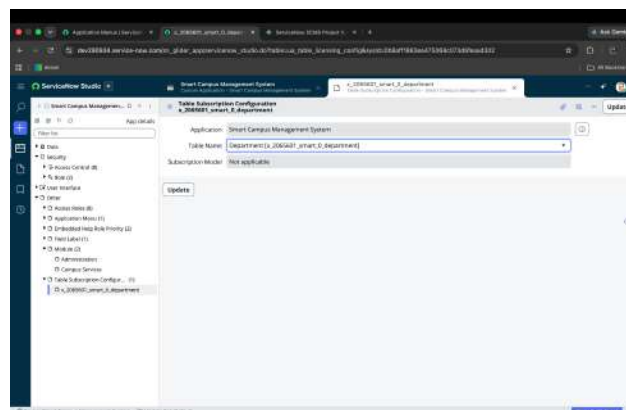


- *Figure 1.3: Module category separators implementation.*

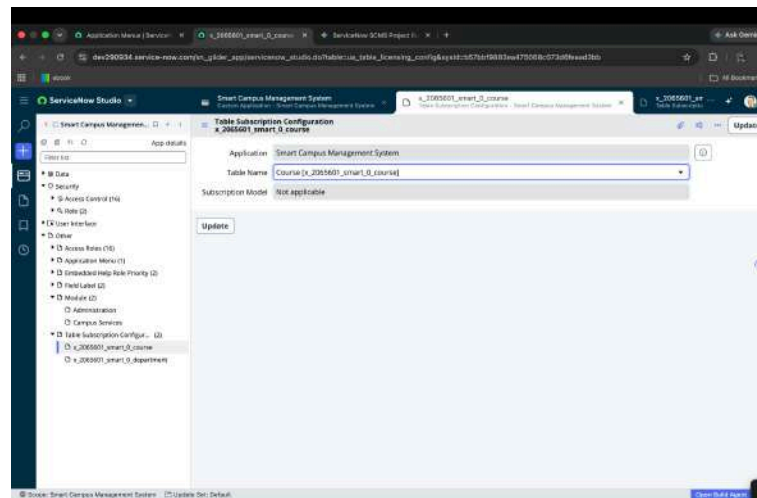


Phase 2: Core Data Schema & Table Architecture

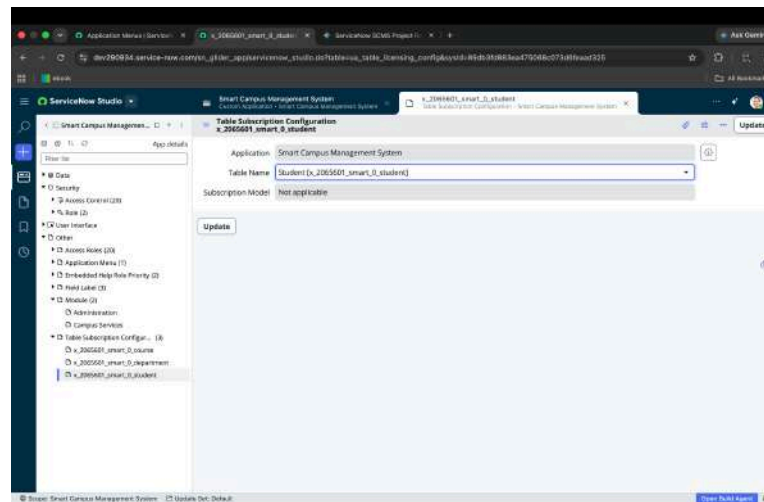
- **Technical Report Matter:** Establishes the underlying database architecture of the platform. A relational schema is deployed utilizing master entity tables for academic departments, courses, students, and faculty, alongside transactional operational tables that log student course registrations, leave requests, and infrastructure maintenance complaints.
 - *Figure 2.1: Department Table schema definition.*



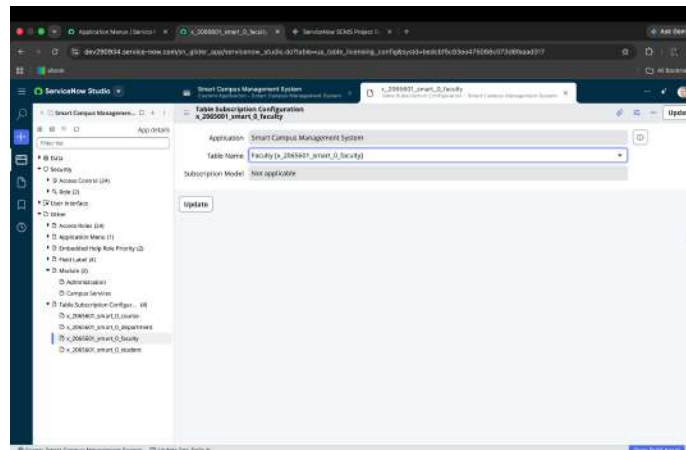
- *Figure 2.2: Course Master Table setup.*



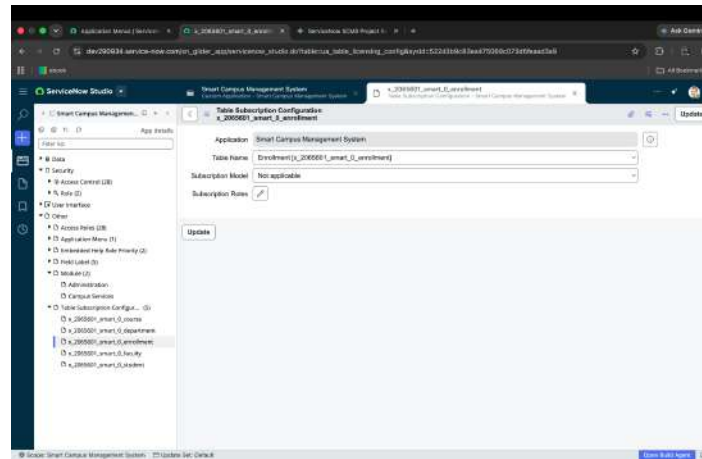
- *Figure 2.3: Student Entity Table configuration.*



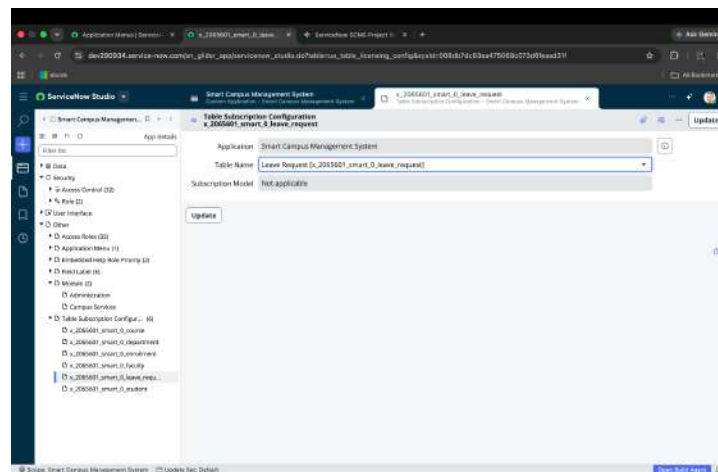
- *Figure 2.4: Faculty Master Table configuration.*



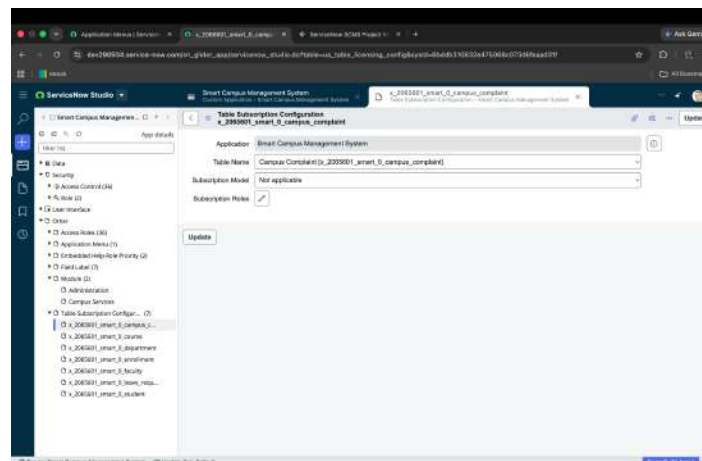
- *Figure 2.5: Student Course Enrollment mapping table.*



- *Figure 2.6: Transactional Leave Request table setup.*



- *Figure 2.7: Campus Facility Complaint table structure.*



Phase 3: Form Layouts, Field Configurations & Dictionaries

- **Technical Report Matter:** Focuses on the fine-tuning of data dictionaries, field attributes, form views, section containers, and list filters across all custom tables. This phase ensures that users and administrators interact with well-structured, intuitive forms containing proper reference qualifiers and choice lists.

- *Figure 3.1: Department dictionary and field setup.*

The screenshot displays the ServiceNow Studio interface for the 'Department' dictionary. The left sidebar shows a tree view of the application components, including 'Department' and 'Course'. The main workspace shows the 'Department' dictionary with fields for 'Name', 'Head of Department', 'Department Code', and 'Status'. The 'Status' field is set to 'Active'. A 'Submit' button is visible at the bottom of the form.

- *Figure 3.2: Course form field specifications.*

The screenshot displays the ServiceNow Studio interface for the 'Course' form. The left sidebar shows a tree view of the application components, including 'Course' and 'Department'. The main workspace shows the 'Course' form with fields for 'Course Code', 'Course Title', 'Credits', and 'Department'. The 'Department' field is set to 'Active'. A 'Submit' button is visible at the bottom of the form.

- *Figure 3.3: Student form field properties.*

The screenshot displays the ServiceNow Studio interface for the 'Student' form. The left sidebar shows a tree view of the application components, including 'Student' and 'Department'. The main workspace shows the 'Student' form with fields for 'Student ID', 'Student Name', 'Department', and 'Academic Year'. The 'Academic Year' field is set to 'First Year'. A 'Submit' button is visible at the bottom of the form.

- *Figure 3.4: Faculty profile field configuration.*

The screenshot shows the 'Faculty Profile' configuration page in ServiceNow Studio. The left sidebar contains a tree view of application components. The main area displays the 'Faculty Profile' form with fields for 'Faculty ID', 'Designation' (set to 'Professor'), 'Faculty Name', 'Status' (set to 'Active'), and 'Department'. A 'Submit' button is visible at the bottom left of the form area.

- *Figure 3.5: Enrollment field dictionary settings.*

The screenshot shows the 'Enrollment' field dictionary settings page in ServiceNow Studio. The left sidebar contains a tree view of application components. The main area displays the 'Enrollment' form with fields for 'Student', 'Enrollment Date', 'Course', 'Status' (set to 'Enrolled'), and 'Academic Term' (set to 'Fall 2028'). A 'Submit' button is visible at the bottom left of the form area.

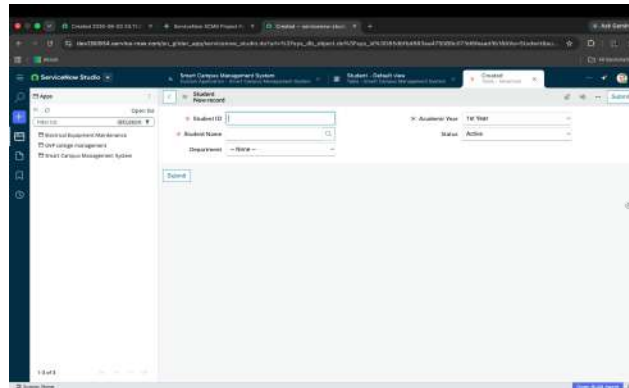
- *Figure 3.6: Leave request form attribute configuration.*

The screenshot shows the 'Leave Request' form attribute configuration page in ServiceNow Studio. The left sidebar contains a tree view of application components. The main area displays the 'Leave Request' form with fields for 'Applicant', 'Leave Type', 'Method', 'Start Date', 'End Date', 'Reason', and 'Approval State' (set to 'Draft'). A 'Submit' button is visible at the bottom left of the form area.

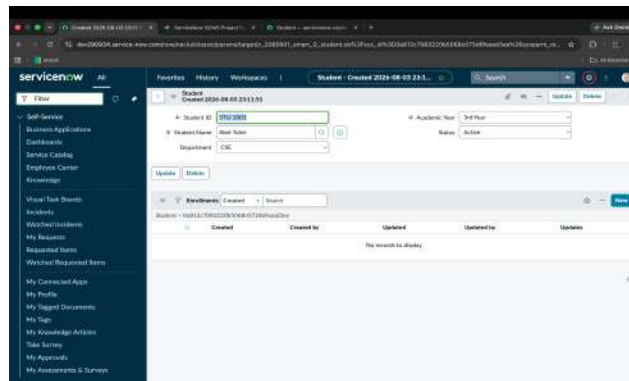
- *Figure 3.7: Complaint tracking field setup.*

The screenshot shows the 'Complaint Tracking' field setup page in ServiceNow Studio. The left sidebar contains a tree view of application components. The main area displays the 'Complaint Tracking' form with fields for 'Category' (set to 'Bugging'), 'Location/Region', 'Reported By', 'Related Department', and 'Resolution Notes'. A 'Submit' button is visible at the bottom left of the form area.

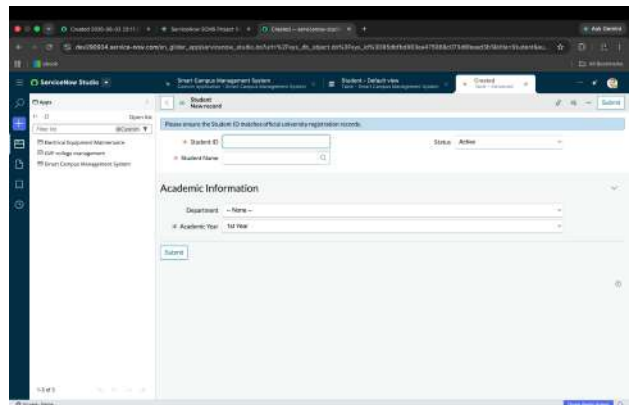
- *Figure 3.8a: Advanced form layout and related lists customization.*



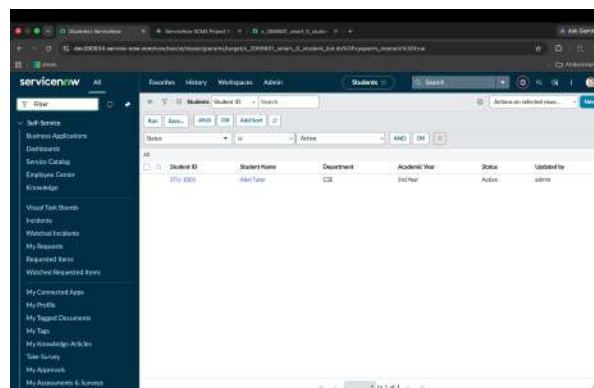
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- *Figure 3.8b: Form view arrangements.*



- Place under form layout design as *Figure 3.9: Form section groupings.*



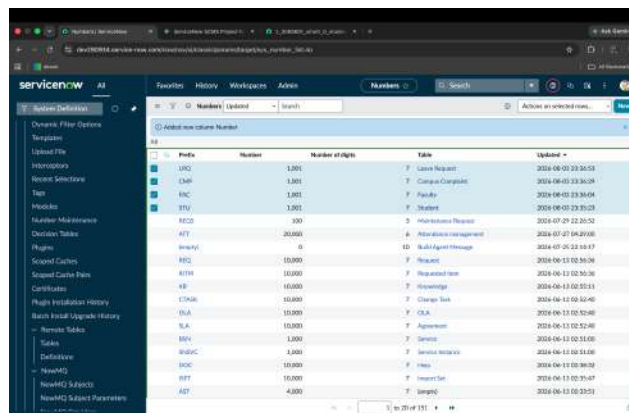
- Place at the conclusion of form configuration as *Figure 3.10: Default list layout and filter configurations.*



Phase 4: Advanced Logic, Automation & UI Policies

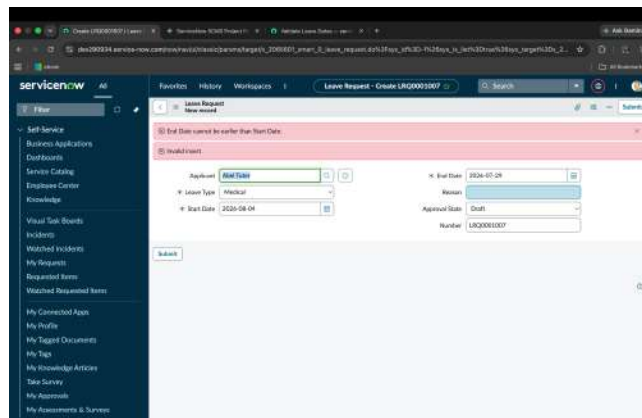
- **Technical Report Matter:** Implements programmatic rules and automation mechanisms to preserve system integrity. Auto-numbering items structure unique record IDs, while custom Client Scripts, UI Policies, and Date Validation scripts enforce real-time visual constraints and input rules. Furthermore, reusable Script Includes ([scmsutils](#)), duplicate prevention rules, and automated ticket routing mechanisms drive backend efficiency.

- *Figure 4.1: Auto-number generator configuration.*

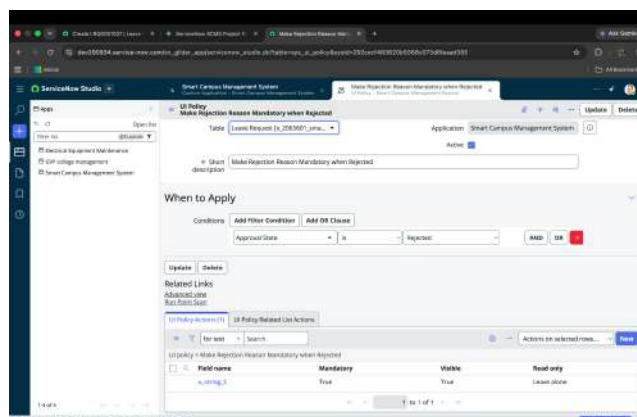


Profile	Number	Number string	Table	Updated
ISC	1,001		F Leave Request	2024-08-01 23:36:53
ENR	1,001		F Campus Complaint	2024-08-01 23:36:28
ENC	1,001		F Faculty	2024-08-01 23:36:04
ETU	1,001		F Student	2024-08-01 23:35:23
STCD	100		S Workforce Request	2024-07-27 22:20:52
MT	20,000		A Absence Management	2024-07-07 04:07:06
HWAG	0		ED Self-Report Message	2024-07-03 23:16:57
REQ	10,000		F Request	2024-06-13 02:56:26
ATTN	10,000		F Requested Item	2024-06-13 02:56:36
KB	10,000		F Knowledge	2024-06-13 02:55:11
CHNG	10,000		F Change Task	2024-06-13 02:52:40
SLA	10,000		F SLA	2024-06-13 02:50:40
SLA	10,000		F Agreement	2024-06-13 02:52:40
SRV	1,000		F Service	2024-06-13 02:51:06
SRVSC	1,000		F Service Request	2024-06-13 02:53:08
PROD	10,000		F Issue	2024-06-13 02:50:20
MTT	10,000		F Incident Set	2024-06-13 02:50:47
ACT	4,000		F Sample	2024-06-13 02:52:53

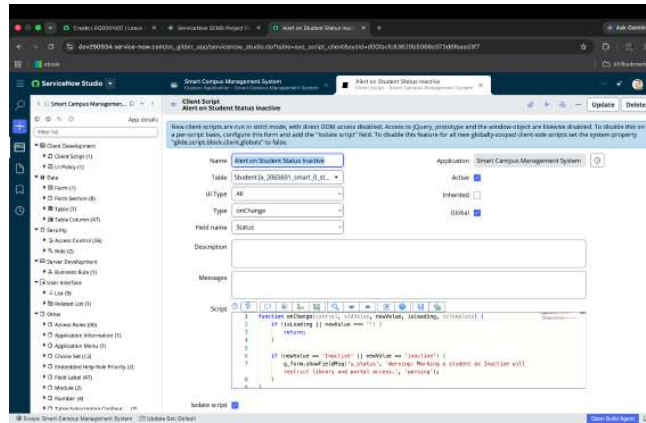
- *Figure 4.2: Core business rule setup.*



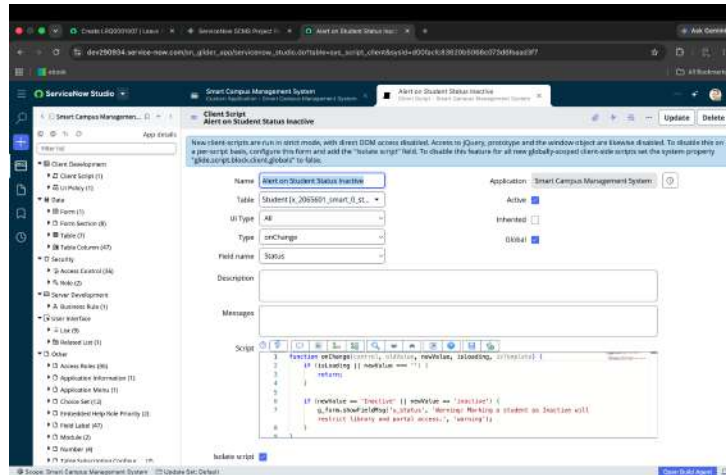
- *Figure 4.3a: Client script framework policies.*



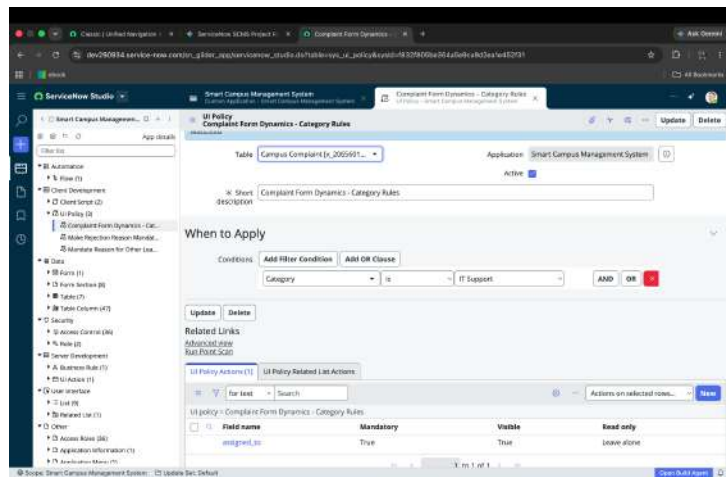
- *Figure 4.3b: Client script behavioral policy.*



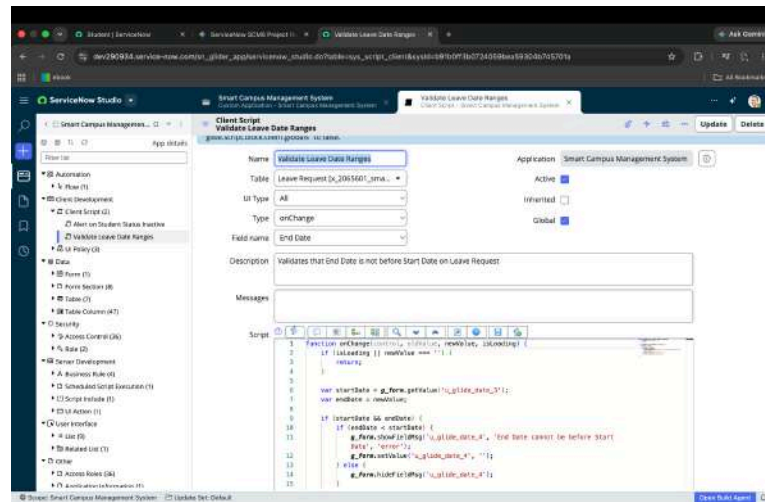
- *Figure 4.3c: Form behavior policy implementation.*



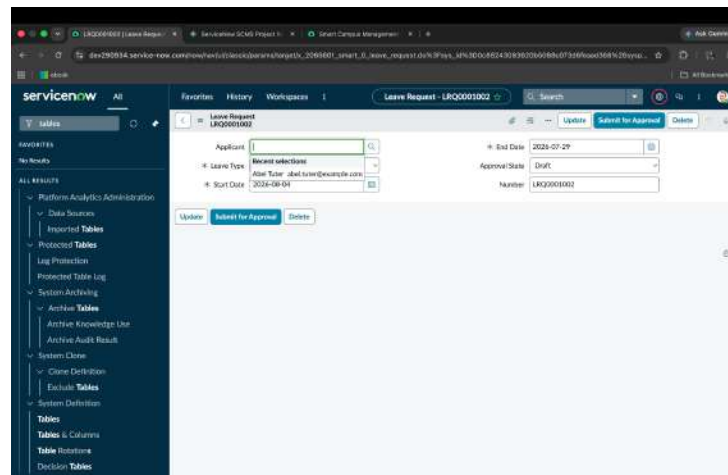
- *Figure 4.4: UI Policy rules for complaint dynamic forms.*



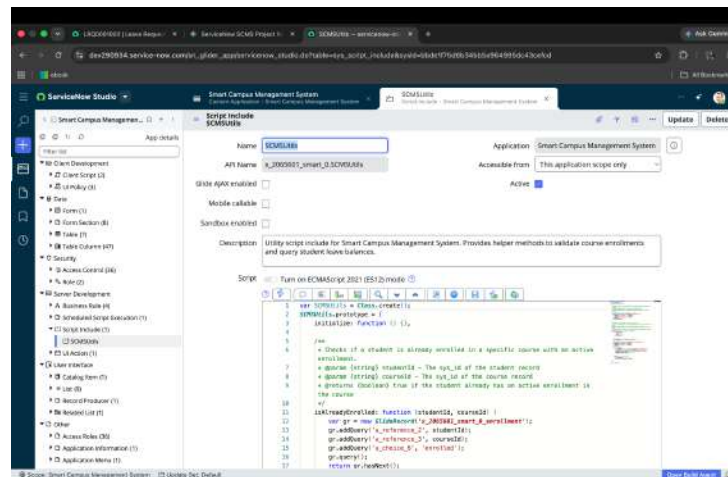
- Figure 4.5: Date validation client script implementation.



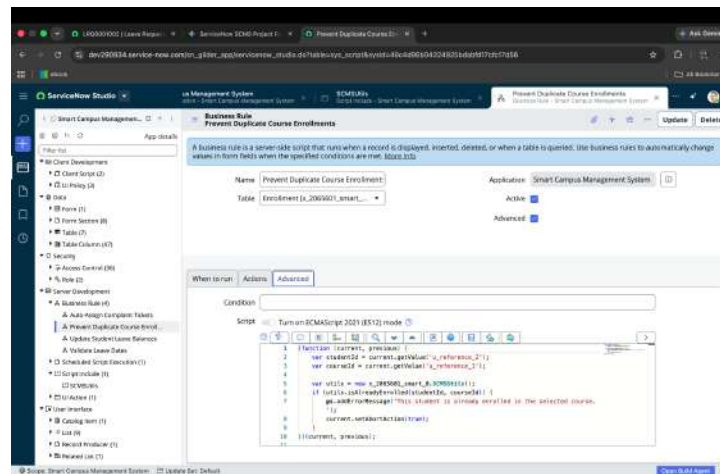
- Figure 4.6: Custom UI Action button configuration.



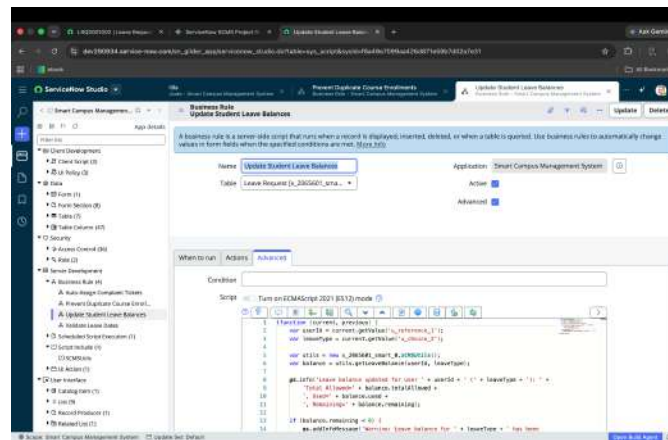
- Figure 4.7: Reusable Script Include utility (scmsutils).



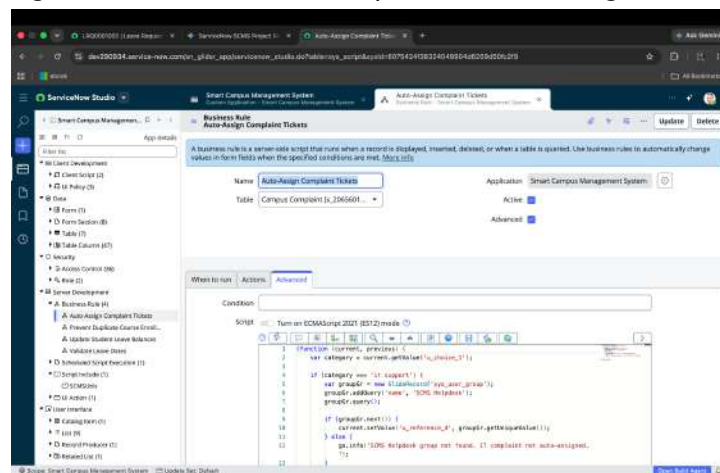
- *Figure 4.8: Duplicate entry prevention business rule.*



- *Figure 4.9a: Student reference updates.*



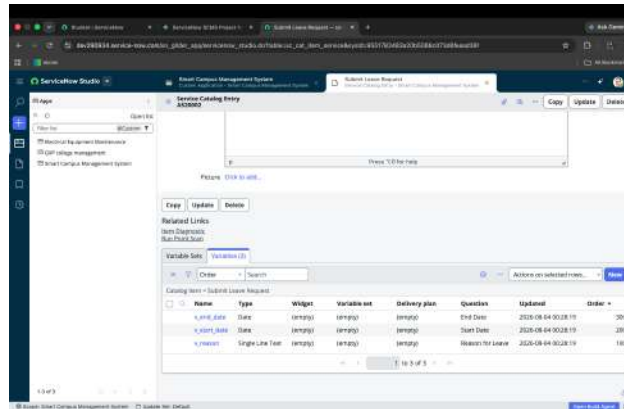
- *Figure 4.9b: Automated complaint ticket routing rules.*



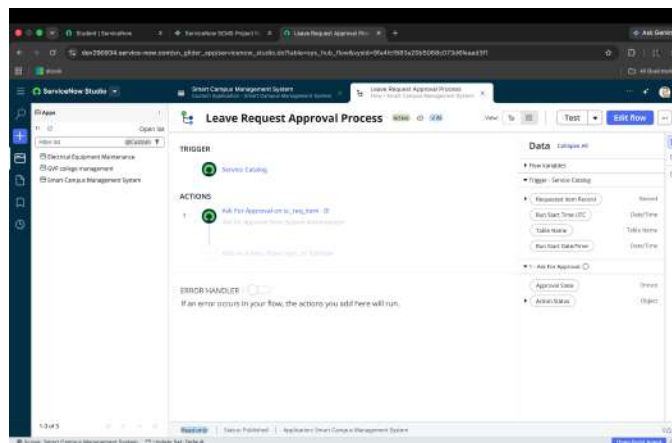
Phase 5: Service Portal, Workflows & Notifications

- **Technical Report Matter:** Bridges backend data structures with the end-user community. Service catalogs and portal interfaces allow students and staff to submit requests smoothly. Behind the scenes, Flow Designer automates the execution path of course registrations and facility complaints, while notification templates trigger email status updates.

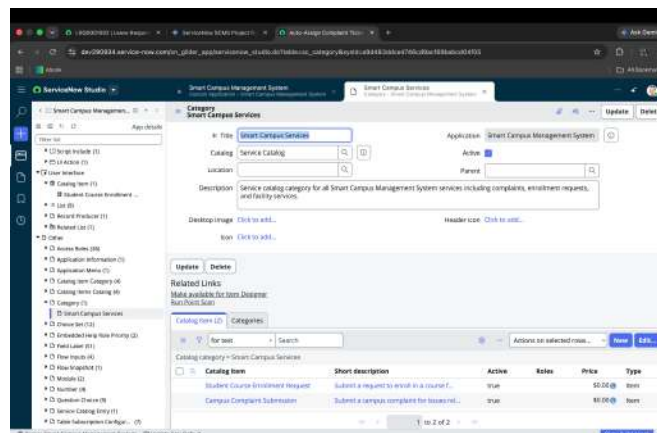
- *Figure 5.1: Service Catalog frontend design.*



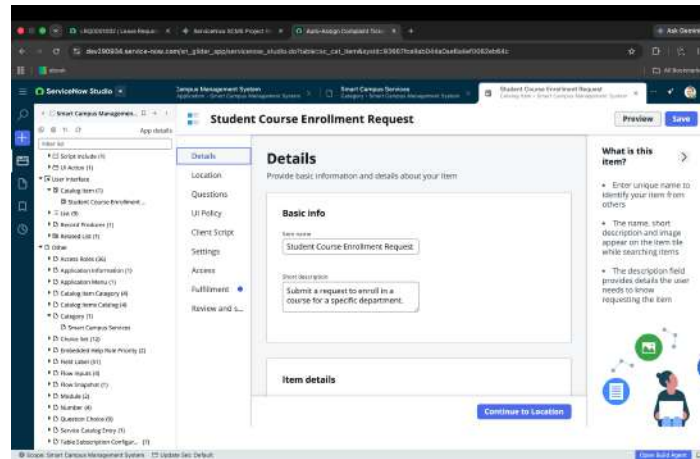
- *Figure 5.2: Catalog item flow designer integration.*



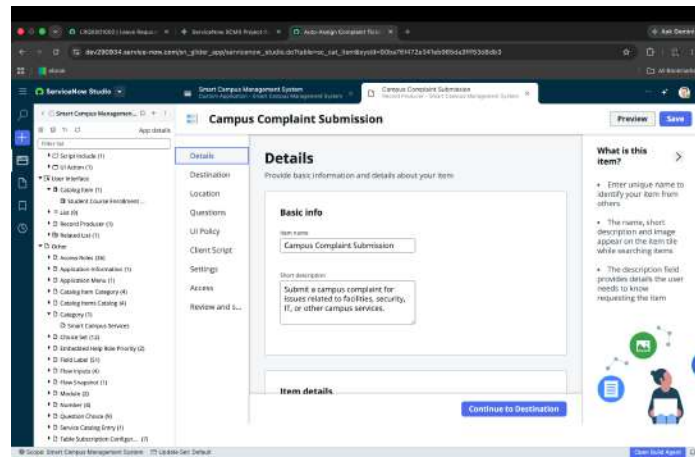
- *Figure 5.3: Smart Campus Services portal landing view.*



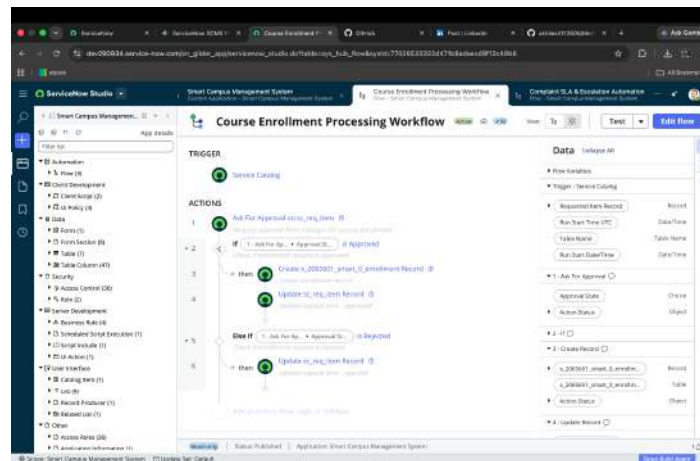
- *Figure 5.4a: Student course request submission interface*



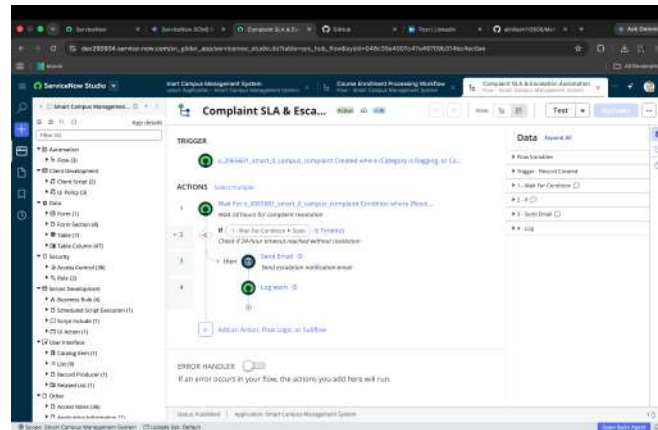
- *Figure 5.4b: Campus complaint portal submission interface.*



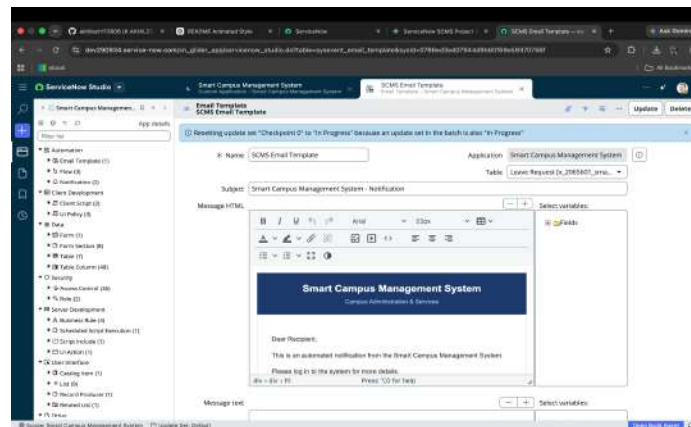
- *Figure 5.5a: Flow Designer sequence for enrollment automation.*



- *Figure 5.5b: Flow Designer sequence for request processing.*



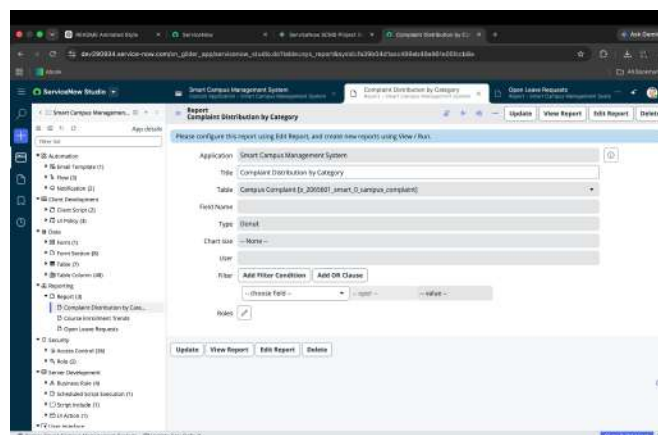
- *Figure 5.6: Leave status email notification template.*



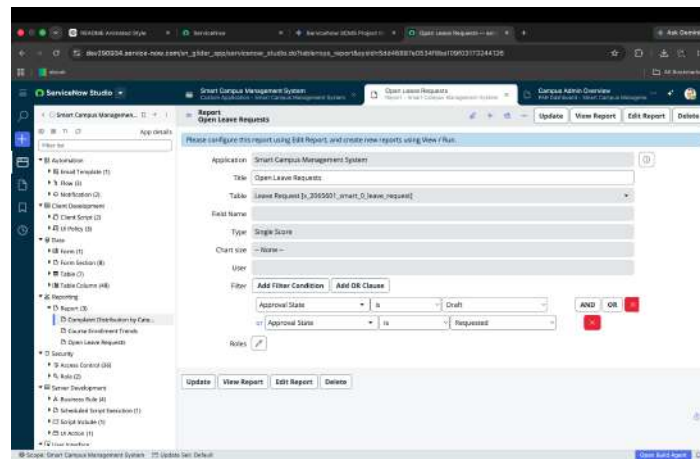
Phase 6: Dashboards, Analytics & System Export

- **Technical Report Matter:** Concludes the system lifecycle by delivering executive visibility and packaging the application. Administrators utilize real-time graphical reporting dashboards to analyze complaint distributions, track open leave backlogs, oversee campus metrics, and review course enrollment trends. Finally, the complete application scope is compiled into an Update Set for deployment.

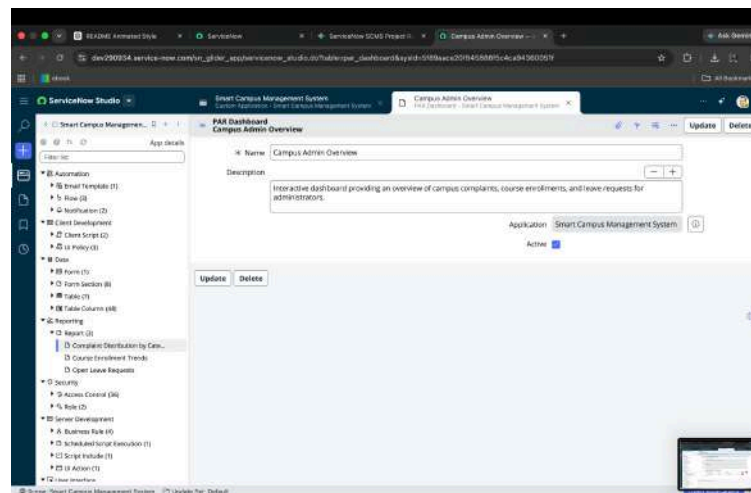
- *Figure 6.1: Complaint distribution analysis by category.*



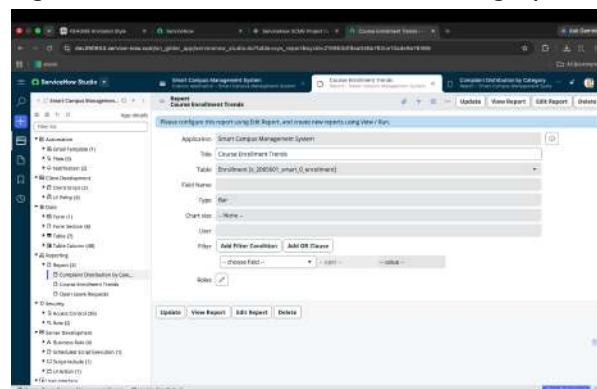
- *Figure 6.2: Open leave request tracking dashboard.*



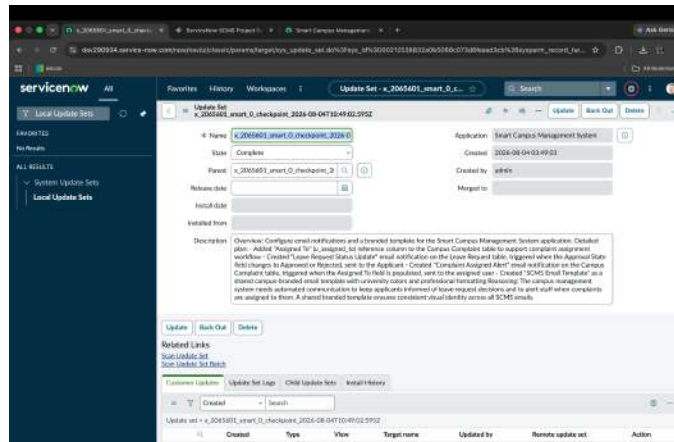
- *Figure 6.3: Campus Administrator high-level overview dashboard.*



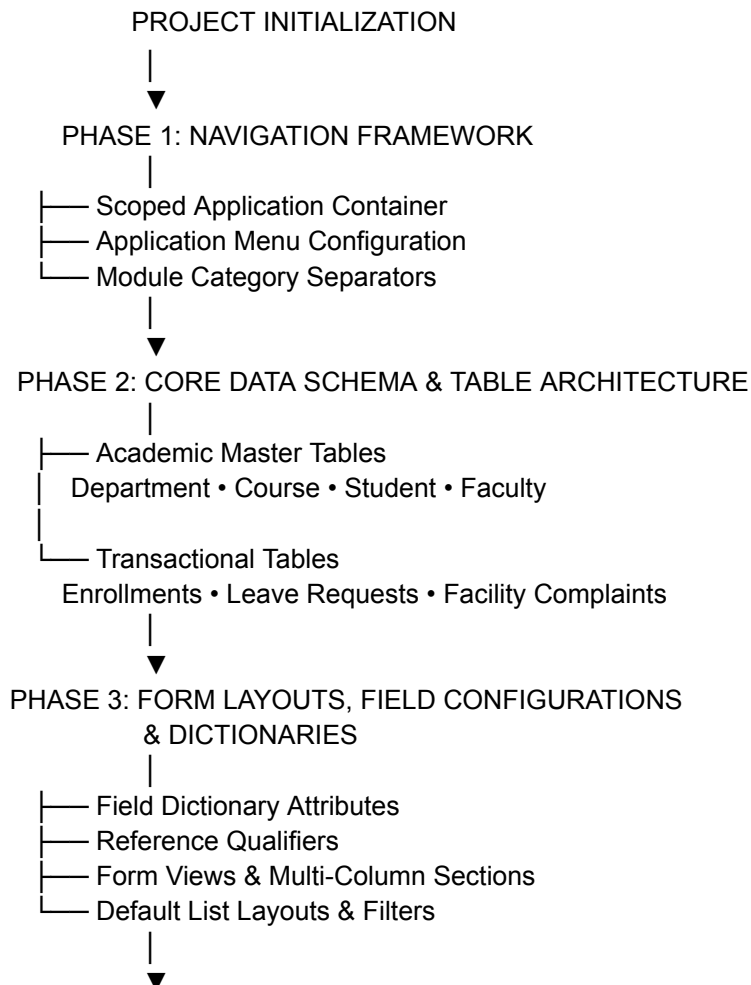
- *Figure 6.4: Course enrollment trends graphical report.*

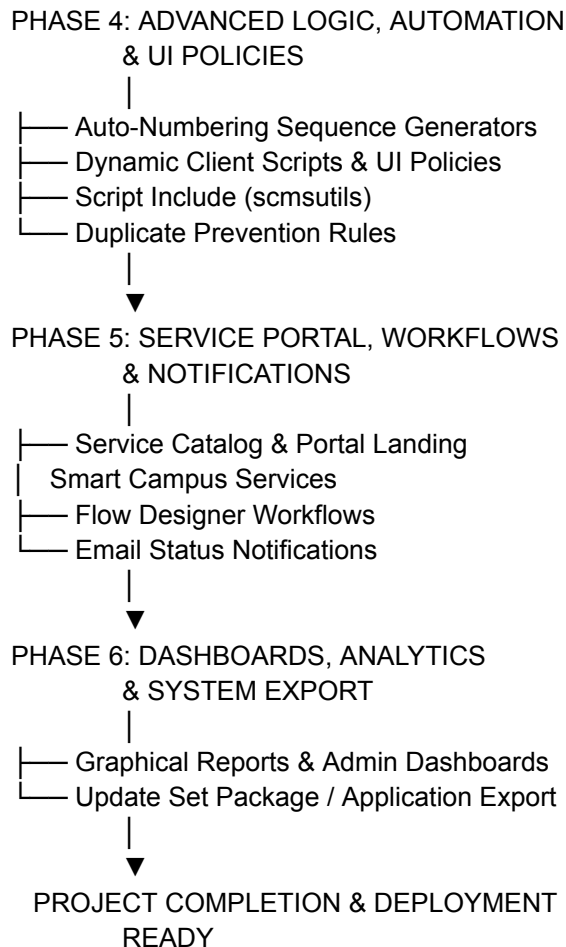


- *Figure 6.5: Final Update Set packaging and export confirmation.*



Smart Campus Management System – Project Flow





Conclusion

The **Smart Campus Management System (SCMS)** successfully demonstrates the power and flexibility of the ServiceNow platform in transforming traditional academic and administrative operations into a streamlined digital experience. By engineering a secure, isolated scoped application (x_2065601_smart_0), the project eliminates fragmented paperwork through robust relational data modeling, automated business logic, real-time client validation, and intuitive self-service portal workflows.

The implementation of Flow Designer automation, dynamic UI policies, and reusable script utilities ensures high data integrity and

efficient task routing. Furthermore, comprehensive reporting dashboards provide leadership with vital operational visibility into course metrics, facility issues, and administrative workloads. Ultimately, the successful packaging of the platform into a centralized Update Set proves its readiness for enterprise deployment, making it a scalable, efficient solution for modern educational institutions.